Atharv Bhale

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Education

University of Massachusetts Amherst — **Amherst, MA**
Master of Science in Computer Science, GPA: 4.0/4.0 — *Expected May 2027*
- Relevant Coursework: Distributed Systems, Software Engineering, Machine Learning, Reinforcement Learning

Savitribai Phule Pune University — **India**
Bachelor of Technology in Computer Engineering, GPA: 3.93/4.0 — *2020 – 2024*

Experience

Amazon Web Services — **East Palo Alto, CA**
Software Development Engineer Intern — *Sep 2026 – Present*
- Developing a **rules engine from scratch for incompatible-parameter prediction across 5 RDS database engines** on the Aurora team, owning technical design, implementation, and validation.
- Compared **3 architectures** and authored the design with principal-engineer and service-owner feedback; built and verified a configuration resolver for memory lookup, formula evaluation, configuration merging, and version checks.
- Deployed and verified a pre-production **AWS CDK/DynamoDB stack** with point-in-time recovery, write-only IAM permissions, cross-account STS access, and CloudWatch alarms.

Tata Consultancy Services — **Pune, India**
Software Developer — *Feb 2025 – Jul 2025*
- Automated diagnostics and reporting workflows with **Python, SQL, and cron**, reducing manual effort by **50%** and saving **8 engineer-hours per week**.
- Cut **PostgreSQL reporting latency from 20s to 14s** using materialized views, indexes, and query redesign; added **CloudWatch alerts across 3 production services**.
- Built **React/SQL internal dashboards** used by **10+ engineers** to investigate SLA metrics, service health, and data-quality issues.

Champhunt Inc. — **Remote**
Product Engineer — *Aug 2024 – Jan 2025*
- Built **React/Node.js subscription and payment workflows** with transaction-status verification, database persistence, and paid-access controls, contributing to **2x recurring revenue over 4 months**.
- Shipped **Reels and Stories** with design and QA using React and Node.js, supporting **40% user growth over 6 months**.
- Reduced infrastructure costs by **25%** by migrating static assets to **AWS S3/CloudFront** and moving asynchronous workloads to **AWS Lambda**.
- Mentored **3 interns** through planning, code reviews, and debugging; coordinated frontend-backend integrations.

Product Engineering Intern — *Jan 2024 – Jul 2024*
- Owned **Fantasy League** data ingestion, scoring logic, leaderboard APIs, and React user flows, increasing user engagement by **45%**.
- Built **Kafka event pipelines** to ingest live match data asynchronously and refresh high-traffic leaderboards after each over.
- Reduced **API response latency by 40%** through **Redis caching**, database indexes, and query refactoring.

Projects

MediConnect | React, Python, FastAPI, Node.js, Socket.IO, MongoDB — **Jan 2026**
- Built a **React/Node.js healthcare platform** with role-based access, appointment scheduling, admin approvals, and real-time chat, integrating AI-assisted clinical review into user workflows.
- Developed **Clinical Copilot**, a **Python/FastAPI service** using LiteLLM/Groq and OpenFDA to normalize clinical files and generate structured medication findings, timelines, risk flags, and traceable reports.

RiskWise | React, Python, FastAPI, tRPC, PostgreSQL, OpenAI API — **May 2026**
- Integrated **GDELT, NASA EONET, and USGS APIs** into a three-agent pipeline for live intelligence, explainable route-risk scoring, and alternative-route recommendations.
- Built a **React operations dashboard** with type-safe tRPC APIs, structured logging, hourly scans, and threshold-triggered webhook alerts to surface actionable findings.

Technical Skills

Languages: Java, Python, C++, TypeScript, JavaScript, SQL
Full-Stack: React, Node.js, Express.js, FastAPI, REST APIs, Redux, Socket.IO, HTML/CSS
Data & Cloud: PostgreSQL, MySQL, MongoDB, Redis, Kafka, AWS (EC2, S3, Lambda, DynamoDB, CloudWatch), Docker
Engineering & AI: Git, Linux, CI/CD, AWS CDK, JMeter, Postman, Cursor, Claude, LiteLLM, OpenAI API

Leadership & Awards

- **Graduate Student Government Senator, UMass Amherst:** Represent **500+ CS graduate students** in university policy discussions.
- Won the **Manus AI Hackathon at Boston Tech Week 2026**, building and shipping a working AI product in under 24 hours.